

ASSIGNMENT NO. 2

Title :

Study and Demonstration of Git and GitHub Commands Using Terminal

1. Aim

To study Git and GitHub and demonstrate commonly used Git commands for creating, managing, tracking, and synchronizing a project repository using the terminal.

2. Introduction

Git is a distributed version control system (DVCS) used to track changes made to files during software development. It allows developers to maintain different versions of a project and return to previous versions whenever required.

Git works locally on a computer, so most Git operations can be performed without an internet connection. Developers can create commits, branches, and view the history of a project locally.

GitHub is a cloud-based platform that provides remote hosting for Git repositories. It allows developers to store their projects online, collaborate with other developers, share source code, and maintain project history.

Git and GitHub are commonly used together. Git manages the version history locally, while GitHub can be used to store and synchronize the repository remotely.

3. Basic Git Concepts

Repository

A repository is a location where Git stores the files and complete version history of a project. A repository can be local or hosted remotely on platforms such as GitHub.

Working Directory

The working directory contains the actual project files on which the developer is currently working.

Staging Area

The staging area is an intermediate area where changes are placed before they are committed. The git add command is used to move changes to the staging area.

Commit

A commit is a saved version of the changes made to a project. Each commit has a unique identifier and a commit message describing the changes.

Branch

A branch is an independent line of development. Branches allow developers to work on new features or modifications without directly affecting the main project.

Remote Repository

A remote repository is a version of the project stored on a remote server such as GitHub. The local repository can be synchronized with the remote repository using commands such as `git push` and `git pull`.

4. Git Workflow

The basic Git workflow can be represented as:

Working Directory → Staging Area → Local Repository → Remote Repository

- Files are created or modified in the Working Directory.
 - `git add` moves the required changes to the Staging Area.
 - `git commit` saves the changes in the Local Repository.
 - `git push` uploads committed changes to the Remote Repository on GitHub.
 - `git pull` downloads the latest changes from the remote repository.
-

5. Software Requirements

- Windows Operating System
 - Git
 - VS Code / Git Bash
 - GitHub Account
 - Internet Connection
-

6. Git and GitHub Commands

1. Check Git Version

```
git --version
```

Purpose: Checks whether Git is installed and displays the installed version.

2. Configure Git Username

```
git config --global user.name "Atharv100906"
```

Purpose: Sets the username that will be associated with Git commits.

3. Configure Git Email

```
git config --global user.email "shinkaratharva06@gmail.com"
```

Purpose: Sets the email address associated with Git commits.

4. Create a New Directory

```
mkdir GitAssignment
```

```
cd GitAssignment
```

Purpose: Creates a new directory for the project and moves into it.

5. Initialize Git Repository

```
git init
```

Purpose: Converts the current project directory into a Git repository.

6. Check Repository Status

```
git status
```

Purpose: Shows the current state of the working directory and staging area.

7. Add Files to Staging Area

`git add .`

Purpose: Adds all modified and untracked files to the staging area.

8. Commit Changes

`git commit -m "Initial commit"`

Purpose: Saves the staged changes as a new version in the local repository.

9. View Commit History

`git log --oneline`

Purpose: Displays the previous commits in a short format.

10. Check Branches

`git branch`

Purpose: Displays the branches available in the repository.

11. Create a New Branch

`git branch feature`

Purpose: Creates a new branch named feature.

12. Switch Branch

`git switch feature`

Purpose: Switches from the current branch to the feature branch.

13. Switch to Main Branch

`git switch main`

Purpose: Switches back to the main branch.

14. Connect Local Repository to GitHub

Purpose: Connects the local repository to the user's remote GitHub repository.

15. Check Remote Repository

`git remote -v`

Purpose: Displays the URL of the remote repository connected to the local repository.

16. Rename Branch to Main

`git branch -M main`

Purpose: Renames the current branch to main.

17. Push Repository to GitHub

`git push -u origin main`

Purpose: Uploads the committed changes from the local repository to the main branch on GitHub.

18. View Changes

`git diff`

Purpose: Displays the differences between the current files and the last committed version.

19. Pull Changes from GitHub

`git pull`

Purpose: Downloads the latest changes from the remote GitHub repository and integrates them into the local repository.

20. Clone a GitHub Repository

Purpose: Creates a local copy of the existing repository hosted on GitHub.

7. Conclusion

Git provides an effective method for version control and helps developers track changes made to a project. GitHub provides remote repository hosting and supports collaboration and sharing of projects. Through this experiment, the basic Git workflow and commonly used Git and GitHub

```
PS G:\VSEC> git --version
git version 2.54.0.windows.1
PS G:\VSEC> █
```

```
PS G:\VSEC> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS G:\VSEC> █
```

```
PS G:\VSEC> git config --global --list
core.editor="C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=Atharv100906
user.email=shinkaratharva06@gmail.com
PS G:\VSEC> █
```

```
PS G:\VSEC> git remote -v
origin https://github.com/Atharv100906/VSEC.git (fetch)
origin https://github.com/Atharv100906/VSEC.git (push)
PS G:\VSEC> █
```

```
PS G:\VSEC> git log --oneline
aa7e694 (HEAD -> main, origin/main) Added Assignment 4
ab57380 Updated assignment files
0a750e9 Added Assignment 1 and Assignment 3
7c73e92 First VSEC assignment
PS G:\VSEC> █
```

```
PS G:\VSEC\Git Commands> git add .
PS G:\VSEC\Git Commands> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   index.html
```

```
PS G:\VSEC\Git Commands> git commit -m "Added Git Commands assignment"
[main fe5aa0a] Added Git Commands assignment
 1 file changed, 12 insertions(+)
 create mode 100644 Git Commands/index.html
PS G:\VSEC\Git Commands> git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
PS G:\VSEC\Git Commands>
```

```
PS G:\VSEC\Git Commands> git diff
diff --git a/Git Commands/index.html b/Git Commands/index.html
index c9a9a74..06b832d 100644
--- a/Git Commands/index.html
+++ b/Git Commands/index.html
@@ -7,6 +7,7 @@
 
     <h1>Git and GitHub Assignment</h1>
     <p>This file is created for demonstrating Git commands.</p>
+    <h2>Learning Git Commands</h2>

</body>
...skipping...
diff --git a/Git Commands/index.html b/Git Commands/index.html
index c9a9a74..06b832d 100644
--- a/Git Commands/index.html
+++ b/Git Commands/index.html
@@ -7,6 +7,7 @@
```

```
PS G:\VSEC\Git Commands> git commit -m "Updated Git Commands HTML file"
[main 42b9914] Updated Git Commands HTML file
 1 file changed, 1 insertion(+)
PS G:\VSEC\Git Commands> git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
PS G:\VSEC\Git Commands>
```

```
PS G:\VSEC\Git Commands> git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 796 bytes | 796.00 KiB/s, done
.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/Atharv100906/VSEC.git
   aa7e694..42b9914  main -> main
PS G:\VSEC\Git Commands>
```

```
PS G:\VSEC\Git Commands> git pull origin main
From https://github.com/Atharv100906/VSEC
 * branch            main            -> FETCH_HEAD
Already up to date.
PS G:\VSEC\Git Commands>
```

```
PS G:\VSEC\Git Commands> cd
PS G:\VSEC\Git Commands> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS G:\VSEC\Git Commands> git branch
* main
PS G:\VSEC\Git Commands> git branch feature
PS G:\VSEC\Git Commands> git branch
feature
* main
PS G:\VSEC\Git Commands> git switch feature
Switched to branch 'feature'
PS G:\VSEC\Git Commands> git branch
* feature
main
PS G:\VSEC\Git Commands>

git status

On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
PS G:\VSEC\Git Commands> git diff
diff --git a/Git Commands/index.html b/Git Commands/index.html
```

```

PS G:\VSEC\Git Commands> git diff
diff --git a/Git Commands/index.html b/Git Commands/index.html
index 06b832d..f5766bf 100644
--- a/Git Commands/index.html
+++ b/Git Commands/index.html
@@ -8,6 +8,8 @@
     <h1>Git and GitHub Assignment</h1>
     <p>This file is created for demonstrating Git commands.</p>
     <h2>Learning Git Commands</h2>
+    <h2>Feature Branch</h2>
+<p>This change is made in the feature branch.</p>

</body>
</html>
\ No newline at end of file
PS G:\VSEC\Git Commands> git add .
PS G:\VSEC\Git Commands> git status
On branch feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   index.html

PS G:\VSEC\Git Commands> git commit -m "Added feature branch changes"
[feature 305fc23] Added feature branch changes
 1 file changed, 2 insertions(+)
PS G:\VSEC\Git Commands> git push -u origin feature
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 12 threads
Compressing objects: 100% (28/28), done.
Writing objects: 100% (33/33), 5.11 KiB | 653.00 KiB/s, done.
Total 33 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), done

```

```
remote: Resolving deltas: 100% (7/7), done.
remote:
remote: Create a pull request for 'feature' on GitHub by visiting:
remote:   https://github.com/Atharv100906/VSEC/pull/new/feature
remote:
To https://github.com/Atharv100906/VSEC.git
* [new branch]      feature -> feature
branch 'feature' set up to track 'origin/feature'.
PS G:\VSEC\Git Commands> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS G:\VSEC\Git Commands> git branch
  feature
* main
PS G:\VSEC\Git Commands> git merge feature
Updating 42b9914..305fc23
Fast-forward
  Git Commands/index.html | 2 ++
  1 file changed, 2 insertions(+)
PS G:\VSEC\Git Commands> git push origin main
To https://github.com/Atharv100906/VSEC.git
```